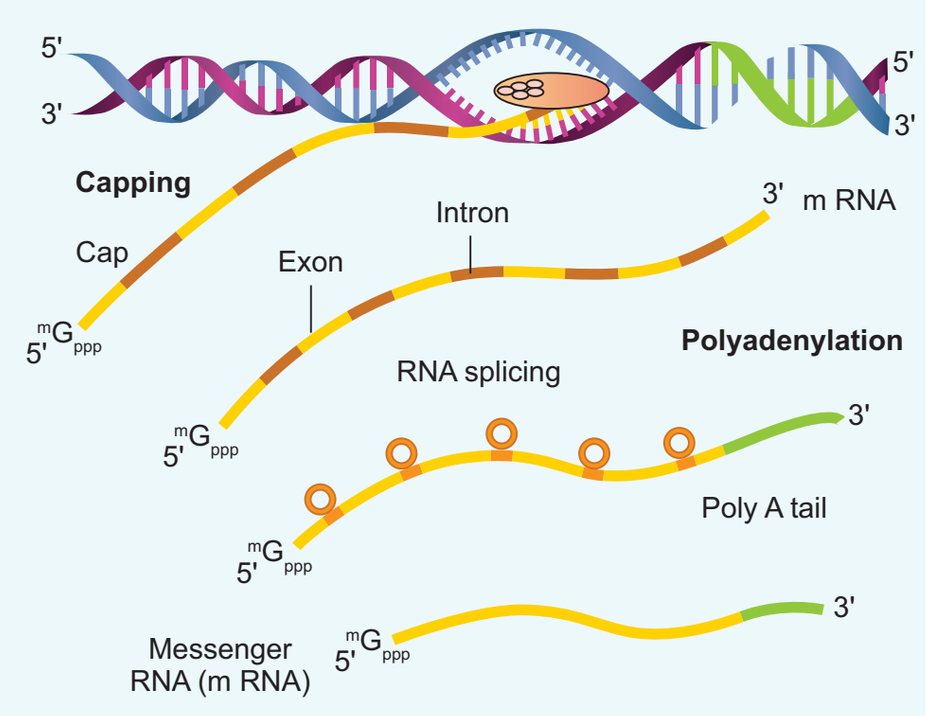


Fig. 5.9 Process of transcription in eukaryotes

patterns, a mechanism known to play an important role in generating both protein and functional diversity in animals. Introns would have arosen before or after the evolution of eukaryotic gene. If introns arose late how did they enter eukaryotic gene? Introns are mobile DNA sequences that can splice themselves out of, as well as into, specific 'target sites' acting like mobile transposon-like elements (that mediate transfer of genes between organisms – Horizontal Gene Transfer - HGT). HGT occurs between lineages of prokaryotic cells, or from prokaryotic to eukaryotic cells and between eukaryotic cells. HGT is now hypothesized to have played a major role in the evolution of life on earth.

5.10 Genetic Code

DNA is the genetic material that carries genetic information in a cell and from generation to generation. At this stage, an attempt will be made to determine in what manner the genetic information exists in DNA molecule? Are they written in coded language on a DNA molecule? If they occur in the language of codes what is the nature of genetic code? The translation of proteins follows the triplet rule; a sequence of three mRNA base (a codon) designates one of the 20 different kinds of amino acids used in protein synthesis.

Genetic code is the sequence relationship between nucleotide in genes (or mRNA) and the amino acids in the proteins they encode. There are 64 possible triplets, and 61 of them are used to represent amino acids. The remaining three triplet codons are termination signals for polypeptide chains. Since there are only 20 amino acids involved in protein synthesis, most of them are encoded by more than one triplet. Two things make this multiple (degenerate) coding possible. First, there is more than one tRNA for most amino acids. Each tRNA has a different anticodon. Second, this pairing is highly specific for the first two portions on the codon, permitting Watson and Crick base pairs (A – U and G - C) to be formed. But at the third position there is a great deal of flexibility as to which base pairs are acceptable. Most part of the genetic code is universal, being the same in prokaryotes and eukaryotes.

The order of base pairs along DNA molecule controls the kind and order of amino acids found in the proteins of an organism. This specific order of base pairs is called genetic code, the blue print establishing the kinds of proteins to be synthesized which makes an organism unique.

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